Rakan Bani-Melhem

Data Scientist

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Professional Summary

Master's-educated Data Scientist with 4 years of expertise in statistical models, Machine Learning, and data visualization. Skilled in Python, tuning supervised and unsupervised Machine Learning models, and building dashboards with analytical storytelling. Proven track record of deploying automated high-performance Machine Learning pipelines and scalable dashboards that improve business performance through data-driven insights. Passionate about transforming complex data into actionable insights that drive smarter decision-making.

Experience & Key Achievements

ScoreFex Intelligent Football Analysis

Data Scientist / Full-Time | Aug 2025 - Oct 2026

- Generated revenue within four months by building automated Machine Learning pipelines for football match outcome prediction.
- Increased customer acquisition by developing and deploying real-time football analytics dashboards that enabled customers to extract key insights effortlessly.
- Substantially reduced financial risk by leading the development of strategic sports odds analytics solutions.

Accuyield SDN BHD

Machine Learning Engineer / Full-Time | Jan 2025 - Jul 2025

- Improved return on investment (ROI) by enhancing predictive model performance through hyperparameter tuning and feature engineering.
- Significantly reduced Machine Learning operational costs and processing time by optimizing end-to-end Machine Learning pipelines.

Data Scientist / Full-Time | Mar 2022 - Dec 2024

- Cut data pipeline operational costs and improved data accuracy by 9% by automating ETL workflows.
- Supported decision-making and business growth by implementing end-to-end Machine Learning pipelines across multiple domains.
- Enhanced operational reliability by implementing monitoring dashboards to continuously track predictive model performance.

Education

MSc in Data Science & Business Analytics / Full-Time | Jan 2021 - Jan 2022

- Dual Degree awarded jointly by Asia Pacific University and De Montfort University, England.

Specializing in Data Science and Machine Learning, with a strong foundation in ETL Pipeline Management, MLOps, and Data Analytics to drive intelligent data-driven solutions and predictive modeling across diverse domains.

Technical & Soft Skills

- **Technical Skills:** scikit-learn, TensorFlow, PyTorch, AWS, Python, SQL, Docker, Airflow, AI Agents, Generative AI.
- **Analysis Skills:** Analysis and Storytelling, Trend Analysis, Analytical Reports, EDA, Tableau, Power BI, Excel.
- **Soft Skills:** Data Driven, Decision Making, Problem Solving, Team Collaboration, Adaptability, Accountability.
- **AI-Assisted Development:** Claude Code, Claude Cowork, Codex, Anthropic API, OpenAI API, MCP.

Projects Experience

Data Integration and ETL Process Management

- Built and managed ETL pipelines that integrated sports data from multiple API sources and web platforms.
- Implemented automated processes to extract, clean, and transform data to ensure quality data for analytics.

Predictive Model Development

- Delivered accurate classification and regression predictions by training and validating supervised ML.
- Developed unsupervised Machine Learning using factor analysis and cluster analysis to reduce data complexity.

Automated Workflows and Model Deployment

- Deployed automated Machine Learning workflows on cloud platforms to ensure scalability and reproducibility.
- Established monitoring systems to track model performance, detect data drift, and trigger retraining workflow.

Data Visualization and Report Analytics

- Developed and deployed user-friendly sports dashboards that allowed users to extract key insights quickly.
- Delivered analytical reports that highlighted critical data analysis findings, model evaluation, and trends.

Forecasting and Trend Analysis

- Developed statistical forecasting models using Machine Learning regression to support marketing strategy.
- Built time series forecasting models using Deep Learning to forecast financial market trends and patterns.

NLP Analytics and LLM Applications

- Applied NLP including sentiment analysis to extract key insights by analyzing reviews and social media data.
- Implemented Retrieval-Augmented Generation (RAG) techniques to enhance model performance by integrating domain-specific knowledge.